

Modification of the Border Detection Algorithm

Motivation for the Modification

The initial border detection algorithm successfully identified the sign language interpreter region in most video frames. However, during experimental evaluation, several limitations were observed. The detected bounding box was often smaller than the actual interpreter window, resulting in a tightly cropped region. Consequently, parts of the interpreter's hands, arms, or surrounding signing space were occasionally excluded from the extracted video. Since sign language recognition relies heavily on complete hand and body movements, losing these visual features could negatively affect the quality of the generated dataset.

Furthermore, the original method primarily depended on detecting bright border colours surrounding the interpreter window. In practical broadcast videos, the border may be thin, partially visible, or contain weak colour variations, reducing the robustness of the detection process.

To overcome these limitations, the border detection algorithm was modified to improve the accuracy and reliability of interpreter region extraction.

Modifications Performed

The following improvements were introduced to the original border detection algorithm:

- Changed the detection strategy from selecting only the inner interpreter region to detecting the complete interpreter window.
- Separated border validation from final bounding-box generation.
- Used the interior region only to verify the presence of a valid interpreter border.
- Combined light-colour border detection with edge-based analysis to improve border identification.
- Introduced an outer-frame expansion mechanism to recover the complete interpreter frame.
- Analysed the left, right, top, and bottom border boundaries independently.
- Implemented automatic estimation of border thickness based on the detected window size.
- Increased the search region around the detected border to recover incomplete or partially visible borders.

- Modified the candidate scoring method to evaluate the complete interpreter region rather than only the detected inner contour.
- Generated a larger and more representative bounding box that preserves the interpreter's complete signing area.
- Reduced the possibility of clipping important hand gestures and body movements during cropping.
- Improved the robustness of the detection process under varying border appearances.

Location of the Modifications

The modifications were implemented in several key sections of the border detection module.

1. Border Detection Function

Function Modified:

`_find_border_in_frame()`

This function was redesigned to improve the interpreter border detection process. Instead of returning only the detected inner region, it now validates the detected border and obtains the complete interpreter frame before generating the final bounding box.

2. Outer Frame Expansion Module

New Function Added:

`_expand_border_box_to_outer_frame()`

A new function was introduced to expand the initially detected border and estimate the complete interpreter window. This module combines colour information with edge information to determine the full outer frame more accurately.

3. Border Validation Strategy

The border validation process was modified by separating the validation stage from the final crop generation. The interior region is now used only to verify whether the detected object represents a valid interpreter border, rather than directly determining the cropping region.

4. Bounding Box Generation

The final bounding-box calculation was modified to use the expanded interpreter frame instead of the smaller interior region. This produces a larger crop that better represents the complete interpreter window.

5. Candidate Evaluation

The scoring mechanism used to select the best interpreter candidate was updated to consider the size of the expanded interpreter region, resulting in more reliable candidate selection.

6. Crop Output Handling

A minor modification was also made to the crop output section to improve crop-adjustment handling and status reporting. This change affects only the output messages and does not alter the detection algorithm itself.

Summary of Improvements

Aspect	Original Algorithm	Modified Algorithm
Detection region	Inner interpreter region	Complete interpreter window
Border validation	Directly used for cropping	Used only for validation
Border information	Light colour detection only	Light colour + edge information
Outer frame detection	Not available	Implemented
Border thickness estimation	Not available	Automatic estimation
Boundary analysis	Limited	Left, right, top, and bottom analysed separately
Candidate scoring	Based on detected contour	Based on expanded interpreter frame
Crop size	Smaller	Larger and more complete
Gesture preservation	Possible clipping	Improved preservation of hand and body movements
Robustness	Moderate	Improved under different border conditions

Figure Reference

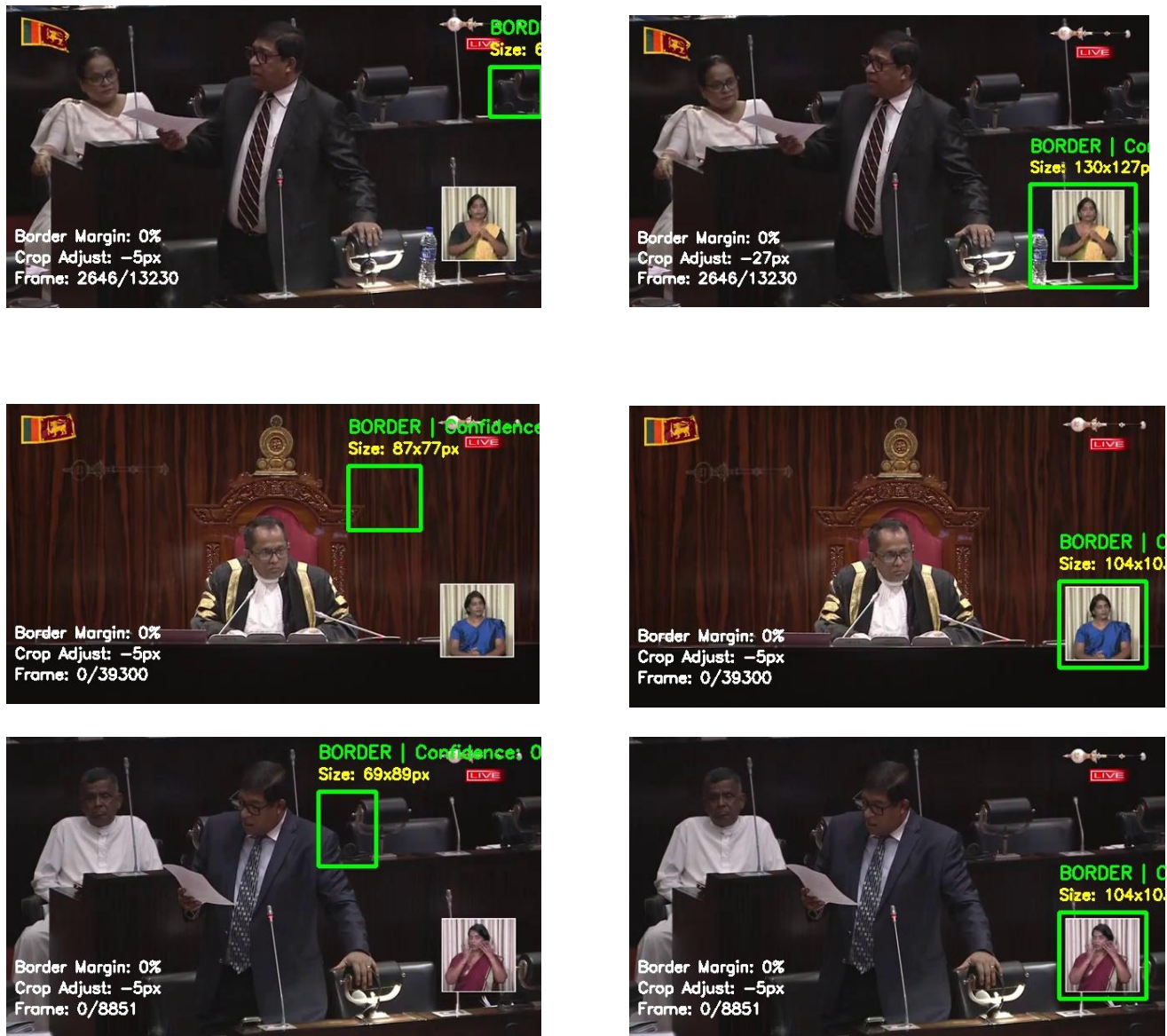


Figure 01: Comparison of the original and modified border detection algorithms. The original algorithm (left) produces a smaller interpreter bounding box, whereas the modified algorithm (right) detects the complete interpreter window by expanding the detected border, resulting in more reliable interpreter extraction